Harness RFQ

Ten low-voltage cable assemblies for a domestic kitchen appliance (a countertop soda dispenser), currently at prototype. Every assembly lands on the main board at one end and on a valve, motor, sensor or display at the other. This document is the complete build package: materials, workmanship, board-side contact order, and a pin-level wire list for each of the ten.

Nothing here is proprietary; ask about anything that is ambiguous rather than assuming.

Scope

In: ten assemblies — J1, J2, J3, J4, J5, J6, J7, J9, J11, J13.

Out: the J10 12 V inlet (2 × 16 AWG, ferrules, landed on the main board) and every AC mains run (AC-1…6, 16 AWG + 18 AWG SJOOW), built in place per `wiring.md`.

Everything in this RFQ is 22 AWG or 28 AWG.

The WAGO fan-outs are built in-house

The five branch nodes in the table below are **excluded from the quote**. A vendor supplies each affected assembly as its trunk only — every board-side conductor run to its full length and terminated at the device end — and the lever nut and its branch legs are made off here. A 221 lever nut takes no crimp and no tooling, and on the one assembly priced both ways the fan-out was 45 % of the assembled cost.

So: quote the wire lists below as written, and ignore the `— … branch` rows.

Build quantity

Quote **25, 50, and 100 sets** (a set = all ten assemblies).

Workmanship standard

IPC/WHMA-A-620, Class 2. Every assembly 100% continuity- and isolation-tested pin-to-pin against its wire list below, including the deliberate non-connection at J2 contact 3.

Materials

Item	Spec	Notes
Signal / actuator wire	22 AWG stranded tinned-copper **silicone**, 600 V, **black**, 1.7 mm ± 0.1 OD	all ten assemblies except J3 — see Open question 1
Ribbon	28 AWG 4-conductor jacketed ribbon, black	J3 only
Board-end housings	JST **XH**, 2.5 mm pitch, female crimp housing + contacts	XHP-*n* per assembly; "XH2.54" is the series name, the pitch is 2.50 mm
Device-end tabs	Female Faston disconnects, **6.3 mm and 4.8 mm**	at 20–24 AWG these are TE **2178438-1** (250 / 6.3 mm) and **170214-2** (187 / 4.8 mm); per-device size map not yet recorded
Screw landings	Insulated bootlace ferrules, DIN, 22 AWG	J5 relay terminals, J9 display terminals
Fan-out splices	**WAGO 221-420** (10-way) and **221-415** (5-way)	**not in scope — we buy and land these ourselves**, see below
Sleeve		both cut ends finished with black heat-shrink

Item	Spec	Notes
	Black PET expandable braided, 1/4" / 1/2" / 3/4" per assembly	
Ties	Black UV-nylon, flush-cut, no proud tail	

Every conductor is black. Identification is **by assembly name at the housing** (heat-shrink marker), not per-conductor. Do not substitute a color code.

Where the trunk ends on the five branched assemblies

Five assemblies terminate a shared rail in a WAGO 221 lever nut at the **device end**, not at the main board: one rail conductor rides the trunk and explodes at the manifold or reservoir. **None of that is in this quote** — you supply the rail conductor cut to the length in its wire list and ferruled, and it ends there. We buy the lever nuts, cut the branch legs, and land them here.

The table is here only so the rail conductor's destination is unambiguous. It is not a build instruction.

Assembly	Nut	Ways used	Feed	Branches
J1 MANIFOLD A	221-420	9 of 10	`COM`	8 valve `+`
J2 MANIFOLD B	221-415	5 of 5	`COM`	V-I, V-J, fan, V-K `+`
J4 SENSORS	221-415	4 of 5	`GND`	1-wire bus, flow meter, moisture plate
J6 REEDS A	221-415	5 of 5	`GND`	4 reeds
J7 REEDS B	221-420	7 of 10	`GND`	6 reeds

Purchased parts, per set

One set = all ten assemblies. Counts are derived from the wire lists below and exclude the fan-outs. Every conductor end appears exactly once: 16 Faston + 13 ferrule + 24 flying = 53, which is the contact count.

#	Part	Spec / MPN	Qty per set	Unit
1	XH housing, 4-way	JST **XHP-4**, female	5	ea
2	XH housing, 5-way	JST **XHP-5**, female	1	ea
3	XH housing, 6-way	JST **XHP-6**, female	1	ea
4	XH housing, 7-way	JST **XHP-7**, female	2	ea
5	XH housing, 9-way	JST **XHP-9**, female	1	ea
6	XH crimp contact	JST **SXH-001T-P0.6**	53	ea
7	Faston 250 receptacle	TE **2178438-1**, 6.3 mm, female, 20–24 AWG	12	ea
8	Faston 187 receptacle	TE **170214-2**, 4.8 mm, female, 20–24 AWG	4	ea
9	Bootlace ferrule, insulated	DIN 46228-4, to suit final gauge	13	ea
10	Hook-up wire	black silicone 600 V, 22 AWG, 1.7 mm ± 0.1 OD	22.7	m
11	Ribbon, 4-conductor	28 AWG jacketed, black	1.0	m
12	Braided sleeve, 3/4"	black PET expandable	0.30	m
13	Braided sleeve, 1/2"	black PET expandable	1.70	m
14	Braided sleeve, 1/4"	black PET expandable	2.55	m
15	Heat-shrink, sleeve ends	black, sized to sleeve	18	ends

#	Part	Spec / MPN	Qty per set	Unit
16	Heat-shrink marker, label	black, one per assembly, printed with the assembly name	10	ea

Wire length by assembly (sum of conductor cut lengths, service loop excluded): J1 3900, J2 2100, J4 3300, J5 600, J6 3000, J7 4200, J9 1600, J11 2400, J13 1600 mm. J3 is the ribbon, quoted by the metre instead.

Items 7 and 8 split on an assumption — see Open question 2. Item 9's barrel is sized for 22 AWG; if Open question 1 moves the gauge, it moves with it.

Not purchased by you: the WAGO 221-415 / 221-420 lever nuts and every branch conductor downstream of them. Those are ours.

Board-end contact order

Read contact order from this table, not from conductor names elsewhere. Two connectors are numbered opposite to how their signals read: J1 runs `OUT8` → `COM`, and J6 runs `GND`, `RA4` … `RA1`. Contact 1 is the pin-1 end of the housing.

Conn.	Housing	1	2	3	4	5	6	7	8	9
J1 MANIFOLD A	XHP-9	`OUT8`	`OUT7`	`OUT6`	`OUT5`	`OUT4`	`OUT3`	`OUT2`	`OUT1`	`COM`
J2 MANIFOLD B	XHP-6	`COM`	`FAN`	**empty**	`OUT3`	`OUT2`	`OUT1`			
J3 FAUCET	XHP-4	`GND`	`V5`	`IO35`	`IO33`					
J4 SENSORS	XHP-7	`3V3`	`GND`	`V5`	`IO25`	`IO26`	`IO27`	`IO23`		
J5 RELAYS	XHP-4	`GND`	`V5`	`IO2`	`IO19`					
J6 REEDS A	XHP-5	`GND`	`RA4`	`RA3`	`RA2`	`RA1`				
J7 REEDS B	XHP-7	`RB1`	`RB2`	`RB3`	`RB4`	`CLO`	`CHI`	`GND`		
J9 DISPLAY	XHP-4	`B`	`A`	`GND`	`V12`					
J11 GAS	XHP-4	`GND`	`V5`	`DOUT`	`AOUT`					
J13 PUMPS	XHP-4	`AM2`	`AM1`	`BM2`	`BM1`					

J2 contact 3 is deliberately empty

J2's wafer is 6-way; only five conductors are populated. `OUT4` is a routed spare that drives no valve. **Crimp five contacts into an XHP-6 in their labelled positions and leave contact 3 empty. Do not close the gap.** Contact 3 is mid-housing, not on an end — a loom filling 1–5 consecutively lands every conductor one position off, which puts the shared 12 V rail on a driver output. Verify the empty cavity at final test.

J4 and J7 share a housing

Both are XHP-7. A swap would put J4's `3V3` / `V5` onto J7's reed inputs. **Label both at the housing.**

Wire lists

Lengths are conductor cut lengths from the main-board contact to the device termination, service loop excluded — add your standard allowance. They are design targets from the enclosure layout, not yet measured against a built enclosure.

J1 — MANIFOLD A · XHP-9 · 3/4" sleeve

Contact	Signal	To	Length	Termination
1–8	`OUT8` … `OUT1`	valves V-H…V-A, one each	~450 mm	female Faston

Contact	Signal	To	Length	Termination
9	`COM`	221-420 at the manifold	~300 mm	ferrule
—	8 × `COM` branch	221-420 → each valve `+`	~150 mm	ferrule / female Faston

Trunk is 300 mm to the manifold, then a 150 mm fan-out per valve. Low-side switching: `COM` carries shared 12 V to every valve `+`; each valve `–` returns on its own `OUT`.

J2 — MANIFOLD B · XHP-6, 5 populated · 1/2" sleeve

Contact	Signal	To	Length	Termination
1	`COM`	221-415 at the manifold	~300 mm	ferrule
2	`FAN`	condenser fan `–`	~400 mm	female Faston
3	—	**empty, do not populate**	—	—
4	`OUT3`	valve V-K `–` at the aft strip	~500 mm	female Faston
5	`OUT2`	valve V-J `–`	~450 mm	female Faston
6	`OUT1`	valve V-I `–`	~450 mm	female Faston
—	4 × `COM` branch	221-415 → V-I, V-J, fan, V-K	150 / 150 / 100 / 200 mm	ferrule / female Faston

All four branches leave the 221-415 at the manifold, 300 mm along the trunk.

J3 — FAUCET · XHP-4 · 28 AWG 4-conductor ribbon

Contact	Signal	To	Length	Termination
1	`GND`	faucet display GND	~1 m	per display, see Open questions
2	`V5`	faucet display 5 V	~1 m	"
3	`IO35`	display TX	~1 m	"
4	`IO33`	display RX	~1 m	"

Straight-through 4-conductor ribbon up the umbilical, no branch. TTL UART; ESD clamping is on the main board, so no cable-end component is required.

J4 — SENSORS · XHP-7 · 1/4" sleeve · three-way branch

Contact	Signal	To	Length	Termination
1	`3V3`	1-wire temp bus	~600 mm	per device
2	`GND`	221-415 on the –X wall aft	~600 mm	ferrule
3	`V5`	flow meter	~150 mm	per device
4	`IO25`	flow meter pulse	~150 mm	per device
5	`IO26`	1-wire data	~600 mm	per device
6	`IO27`	moisture sensor DO	~600 mm	per device
7	`IO23`	moisture sensor switched VCC	~600 mm	per device
—	3 × `GND` branch	221-415 → 1-wire, flow meter, moisture	short pigtails	ferrule

The flow-meter leg (contacts 3, 4) leaves the trunk at ~150 mm; the other two legs run on to ~600 mm.

J5 — RELAYS · XHP-4 · 1/4" sleeve

Contact	Signal	To	Length	Termination
1	`GND`	relay #1 and #2 GND	~150 mm	ferrule, teed to both
2	`V5`	relay #1 and #2 VCC	~150 mm	ferrule, teed to both
3	`IO2`	relay #2 IN	~150 mm	ferrule
4	`IO19`	relay #1 IN	~150 mm	ferrule

`V5` and `GND` tee at the relay screw terminals — **no WAGO on this assembly.**

J6 — REEDS A · XHP-5 · 1/4" sleeve

Contact	Signal	To	Length	Termination
1	`GND`	221-415 at reservoir A	~600 mm	ferrule
2–5	`RA4 … RA1`	reservoir A level reeds 4…1	~600 mm	reed leads
—	4 × `GND` branch	221-415 → each reed	short pigtails	ferrule

Note the reversed order: contact 2 is `RA4`, contact 5 is `RA1`.

J7 — REEDS B · XHP-7 · 1/4" sleeve

Contact	Signal	To	Length	Termination
1–4	`RB1 … RB4`	reservoir B level reeds 1…4	~600 mm	reed leads
5	`CLO`	carbonator low reed	~600 mm	reed leads
6	`CHI`	carbonator high reed	~600 mm	reed leads
7	`GND`	221-420 at the cold-core end	~600 mm	ferrule
—	6 × `GND` branch	221-420 → each reed	short pigtails	ferrule

Same XHP-7 as J4 — label at the housing.

J9 — DISPLAY · XHP-4 · 1/2" sleeve

Contact	Signal	To	Length	Termination
1	`B`	display RS485 B	~400 mm	ferrule
2	`A`	display RS485 A	~400 mm	ferrule
3	`GND`	display 7–36 V input GND	~400 mm	ferrule
4	`V12`	display 7–36 V input +	~400 mm	ferrule

`A` and `B` twisted pair over the run; `V12` / `GND` ride the same sleeve.

J11 — GAS · XHP-4 · 1/4" sleeve

Contact	Signal	To	Length	Termination
1	`GND`	MQ-6 GND	~600 mm	per device
2	`V5`	MQ-6 VCC	~600 mm	per device
3	`DOUT`	MQ-6 digital trip	~600 mm	per device

Contact	Signal	To	Length	Termination
4	`AOUT`	MQ-6 analog	~600 mm	per device

Straight-through, no branch.

J13 — PUMPS · XHP-4 · 1/2" sleeve

Contact	Signal	To	Length	Termination
1	`AM2`	pump A motor tab 2	~400 mm	female Faston
2	`AM1`	pump A motor tab 1	~400 mm	female Faston
3	`BM2`	pump B motor tab 2	~400 mm	female Faston
4	`BM1`	pump B motor tab 1	~400 mm	female Faston

Two differential H-bridge pairs, no shared rail and no fan-out. The pumps ride the pump cartridge, which withdraws from the enclosure, so these four run **unbroken** from housing to motor tab, and the length must hold with that cartridge drawn fully out — do not shorten.

Open questions for the vendor

1. **Wire gauge into the XH housings — 22 AWG silicone, and we want your read.** JST rates the XH contact (SXH-001T-P0.6) #28 to #22 on the conductor and **0.9 to 1.9 mm on the insulation OD**. The silicone we run measures 1.7 mm ± 0.1 at 22 AWG, so it is inside that window with roughly 0.1 mm to spare at the top of the band. Some crimp-selection databases refuse the pair anyway on their own margins, accepting 22 AWG into XH on UL 1007, UL 1015, UL 1061 and ThermoThin and silicone only from 24 AWG down. **Quote 22 AWG silicone. If your process data will not accept it, say so and quote what you would build instead** — 24 AWG silicone, or 22 AWG thin-wall. Current draw on these looms is negligible except MANIFOLD A `COM`, which carries ≤ ~1.4 A over ~300 mm.
2. **Faston sizes.** `cable-assemblies.md` calls for both 6.3 mm and 4.8 mm disconnects; the per-device map is not yet recorded. Quote against a stated assumption and we will confirm before release.
3. **Device-end terminations on J3, J4, J11.** Sensor and display leads land per device rather than on a standard terminal. Quote these as flying leads, tinned and ferruled, unless you would rather we specify a mating connector.
4. **28 AWG ribbon.** Confirm you can crimp XH contacts onto 28 AWG, or propose the gauge you would rather run in the umbilical.

What we need back

1. **Unit price at 25, 50 and 100 sets**, and any NRE / setup charged per assembly design.
2. **Lead time** for the first article and for a production run.
3. **Your answer to Open question 1** — the wire gauge into the JST XH housings. This is the one open engineering item and we would rather take your recommendation than impose ours.
4. Whether you will build to **IPC/WHMA-A-620 Class 2** and provide continuity + isolation test records per assembly.
5. Anything in the wire lists you would change to make the part cheaper or more manufacturable.

We are quoting several suppliers on the identical package.